Task 1 (AWS Cafe Challenge Portfolio)

Challenge 5 - Creating a Scalable and Highly Available Environment for the Café

Academic Year	Module	Assessment Number	Assessment Type
2026	5CS048 - Cloud Systems	Task 1 Challenge 5	Portfolio

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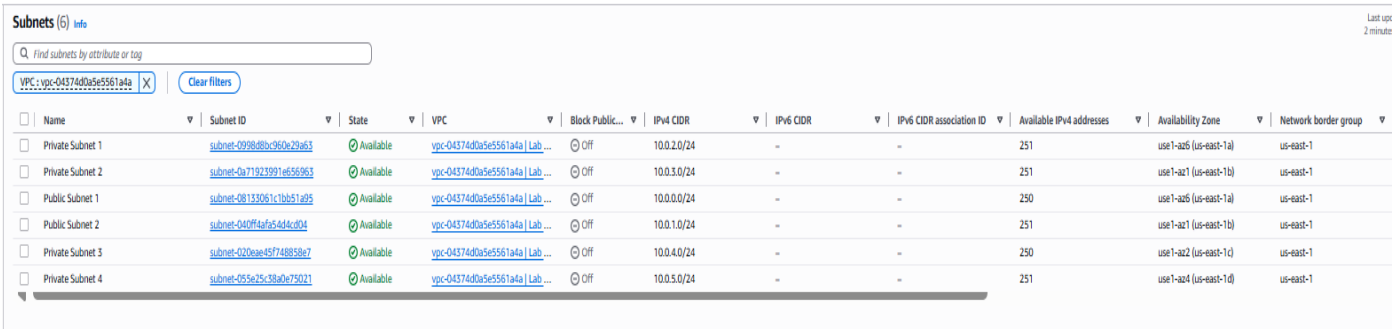
Challenge (Cafe) Lab: Creating a Scalable and Highly Available Environment for the Café

Task 1: Inspecting your environment

Show a screenshot of the Amazon VPC console displaying the Lab VPC details, including:

- **The list of subnets (Public Subnet 1, Public Subnet 2, Private Subnet 1, Private Subnet 2) with their Availability Zones visible.**



Subnets (6) Info

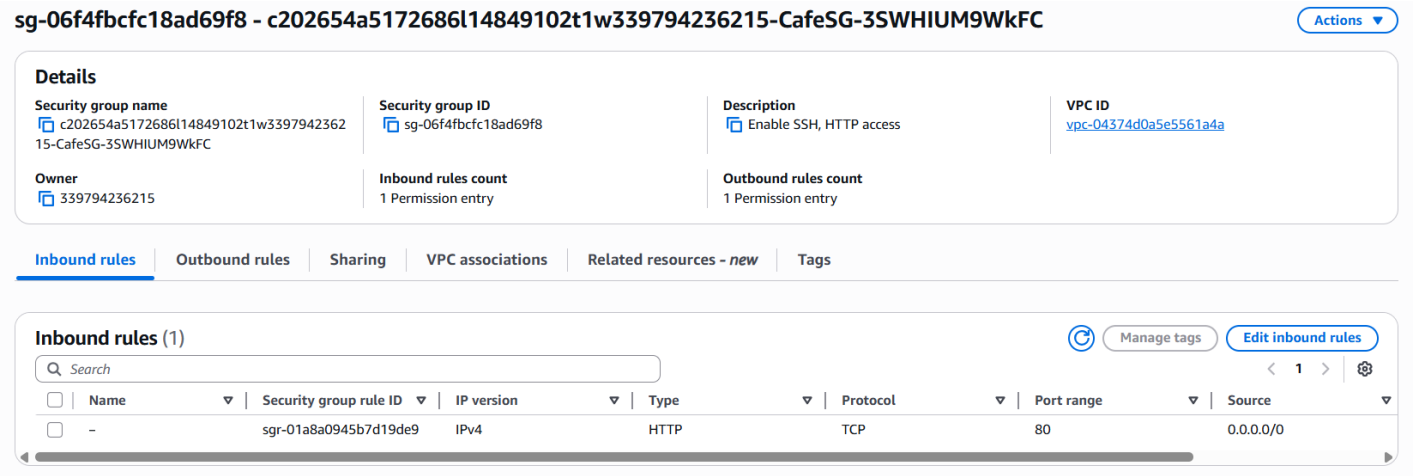
Last upd
2 minute

Find subnets by attribute or tag

VPC : vpc-04374d0a5e5561a4a | Clear filters

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association ID	Available IPv4 addresses	Availability Zone	Network border group
Private Subnet 1	subnet-0998d8bc960e29a63	Available	vpc-04374d0a5e5561a4a | Lab ...	Off	10.0.2.0/24	-	-	251	use1-az6 (us-east-1a)	us-east-1
Private Subnet 2	subnet-0a71923991e656963	Available	vpc-04374d0a5e5561a4a | Lab ...	Off	10.0.3.0/24	-	-	251	use1-az1 (us-east-1b)	us-east-1
Public Subnet 1	subnet-08133061c1bb51a95	Available	vpc-04374d0a5e5561a4a | Lab ...	Off	10.0.0.0/24	-	-	250	use1-az6 (us-east-1a)	us-east-1
Public Subnet 2	subnet-040ff4afa54d4cd04	Available	vpc-04374d0a5e5561a4a | Lab ...	Off	10.0.1.0/24	-	-	251	use1-az1 (us-east-1b)	us-east-1
Private Subnet 3	subnet-020eae45f748858e7	Available	vpc-04374d0a5e5561a4a | Lab ...	Off	10.0.4.0/24	-	-	250	use1-az2 (us-east-1c)	us-east-1
Private Subnet 4	subnet-055e25c38a0e75021	Available	vpc-04374d0a5e5561a4a | Lab ...	Off	10.0.5.0/24	-	-	251	use1-az4 (us-east-1d)	us-east-1

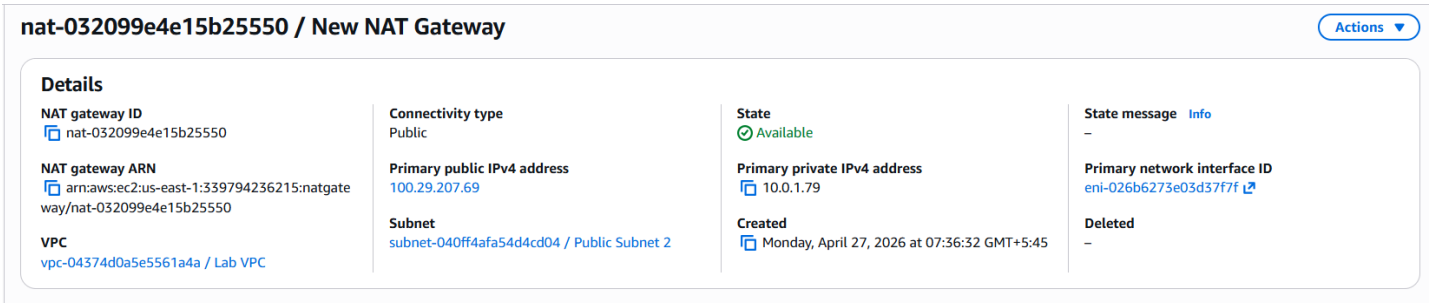
- **The CafeSG security group inbound rules showing which ports are open.**



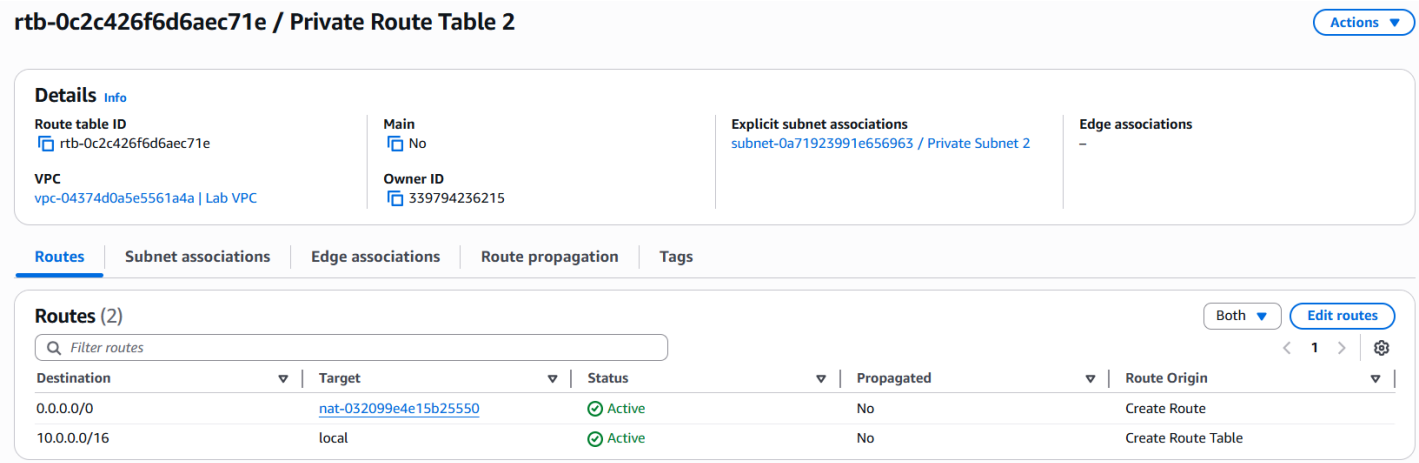
Task 2: Updating a network to work across multiple Availability Zones

Show two screenshots:

- **The VPC console NAT Gateways list showing the newly created NAT gateway in Public Subnet 2 with status Available.**



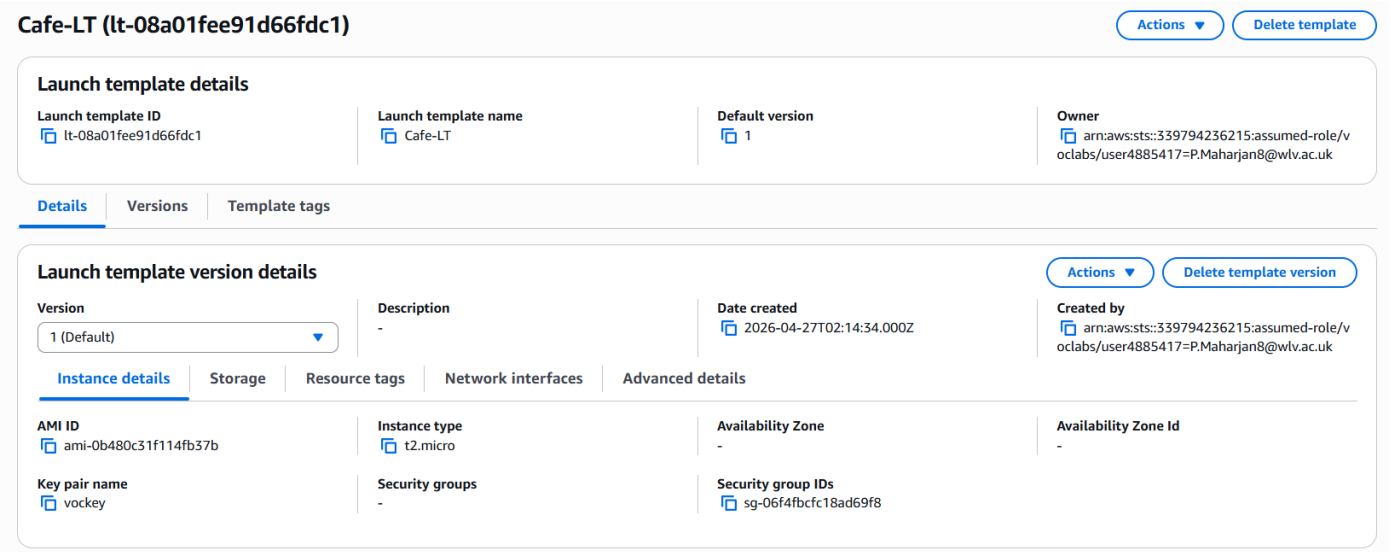
- **The route table associated with Private Subnet 2, showing a route for 0.0.0.0/0 pointing to the new NAT gateway.**



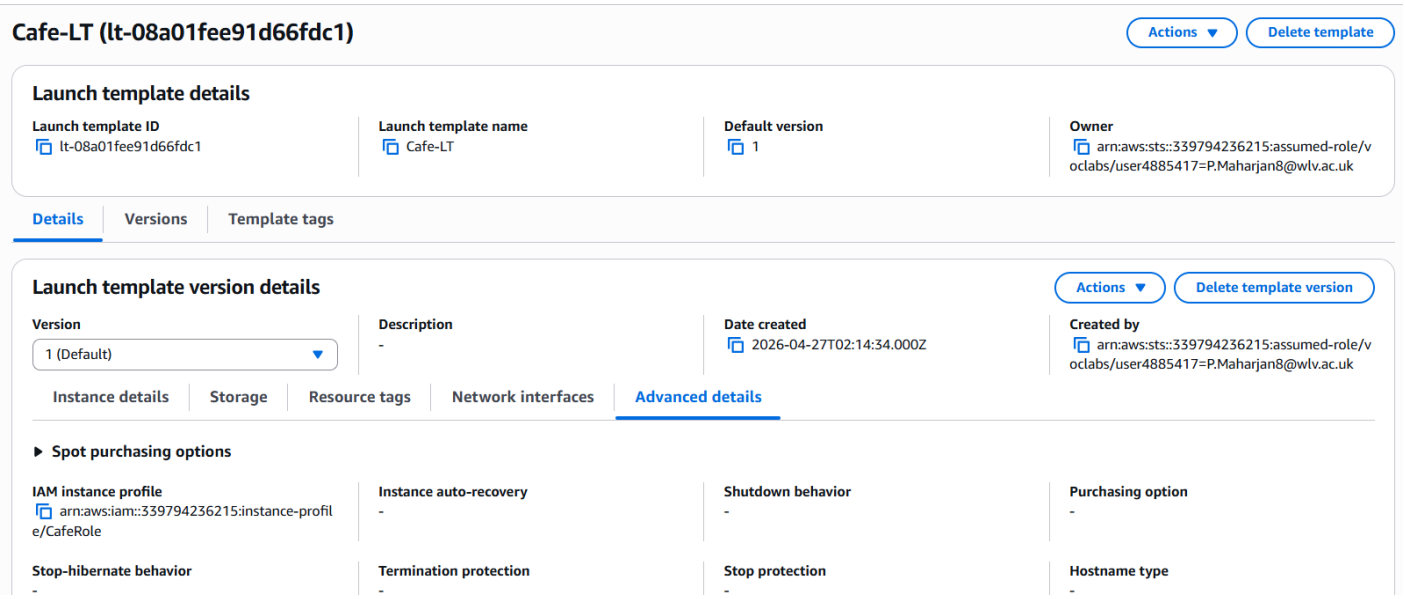
Task 3: Creating a launch template

Show a screenshot of the EC2 Launch Templates console displaying the details of your newly created launch template, with the following visible:

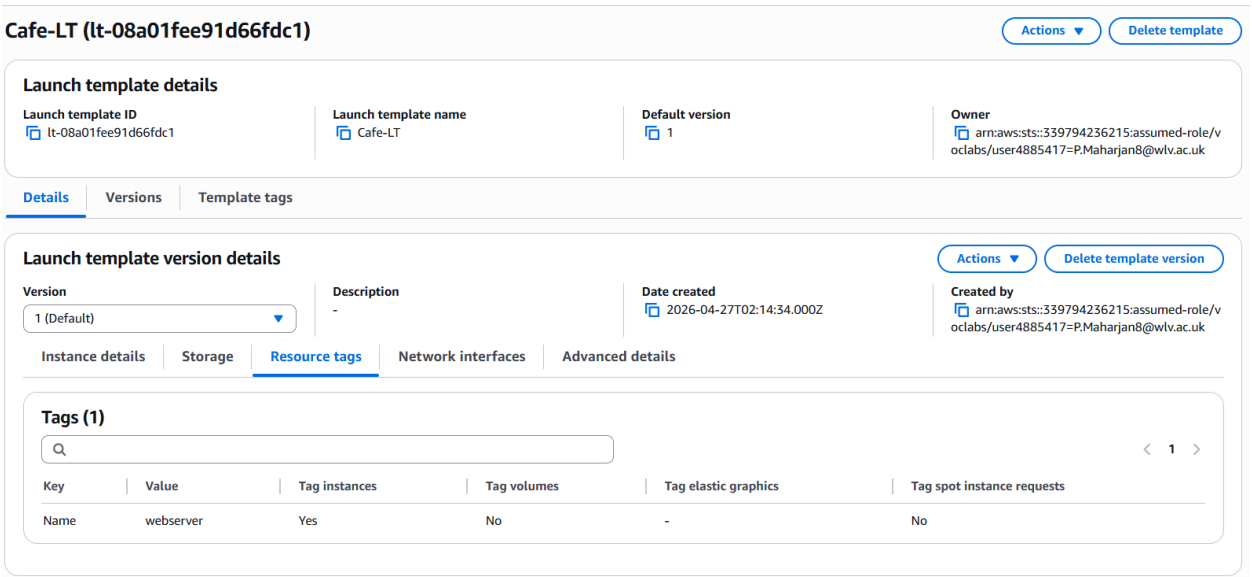
- **AMI: Cafe WebServer Image**
- **Instance type: t2.micro**
- **Security group: CafeSG**



- **IAM instance profile: CafeRole (visible in Advanced details)**



- **Resource tag: Name = webserver**

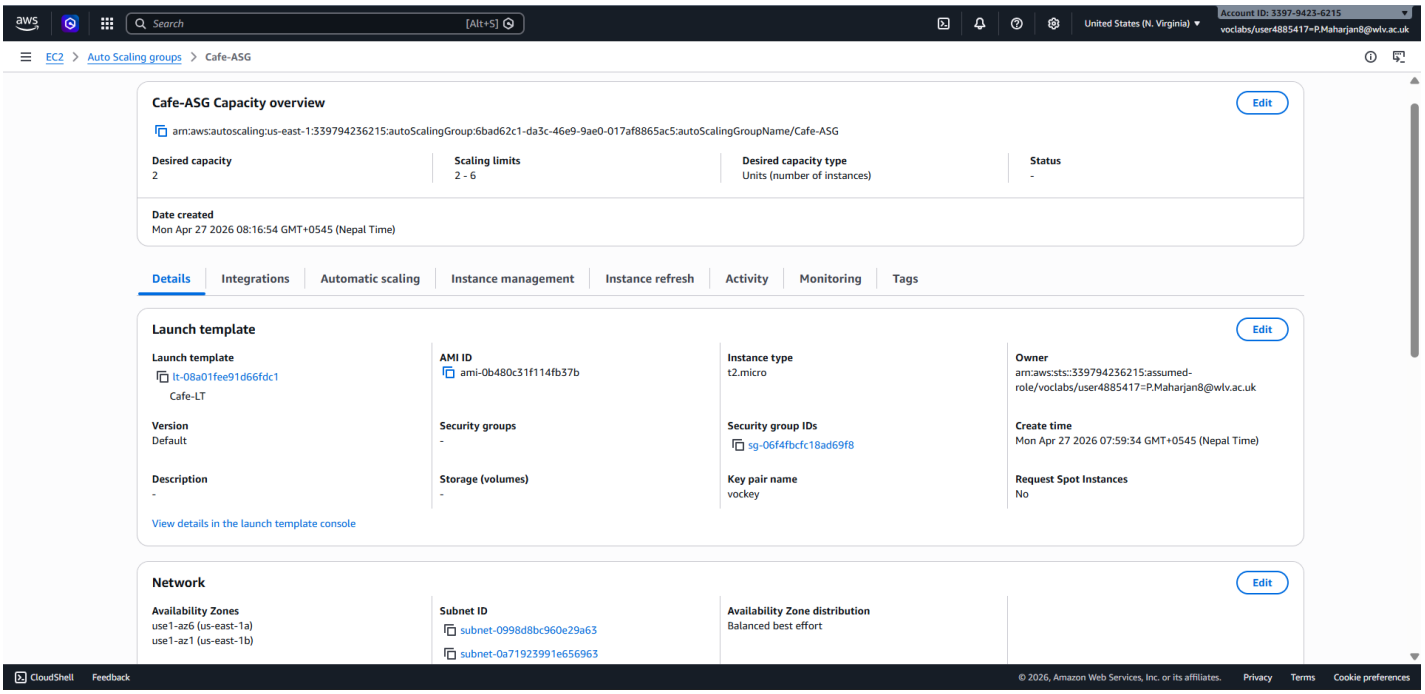


Task 4: Creating an Auto Scaling gro up

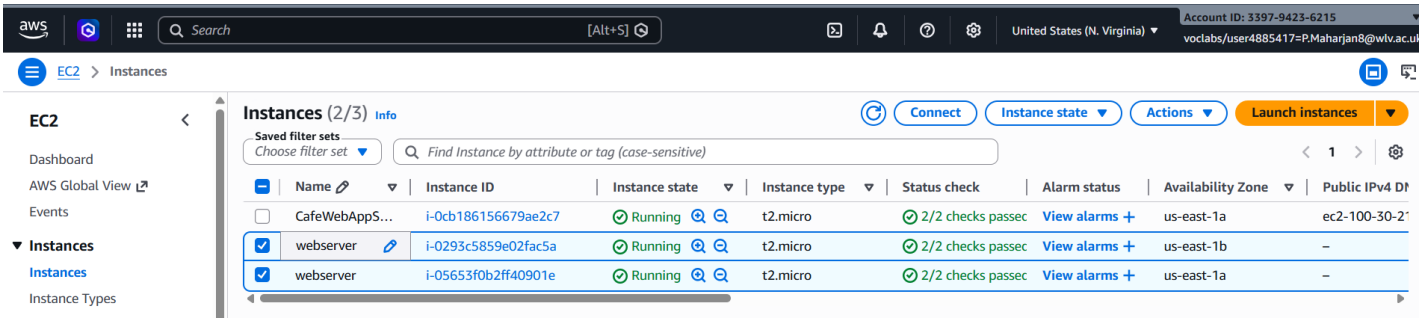
Show two screenshots:

- **The Auto Scaling group details page showing: the launch template selected, VPC/subnets (Private Subnet 1 and Private Subnet 2), and**

group size (Desired: 2, Min: 2, Max: 6).



- The EC2 Instances console listing the two running webserver instances launched by the Auto Scaling group, each tagged with the name webserver.

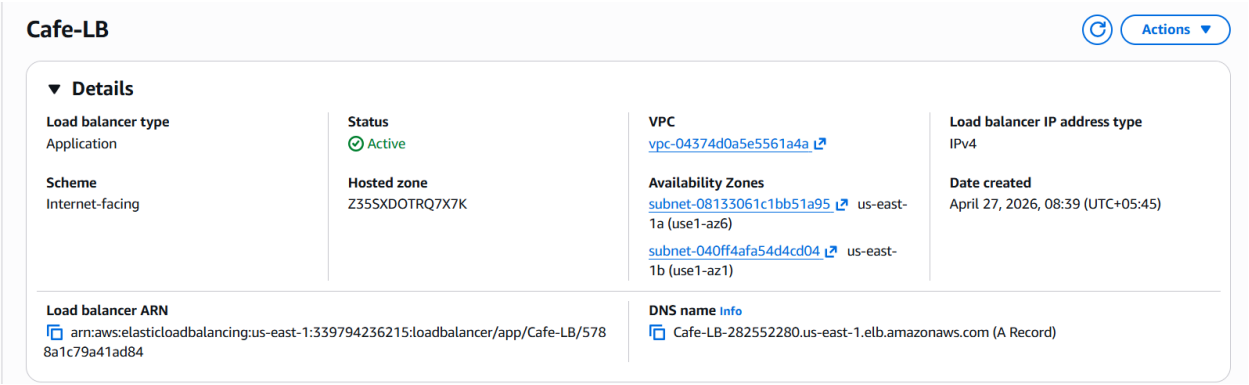


Task 5: Creating a load balancer

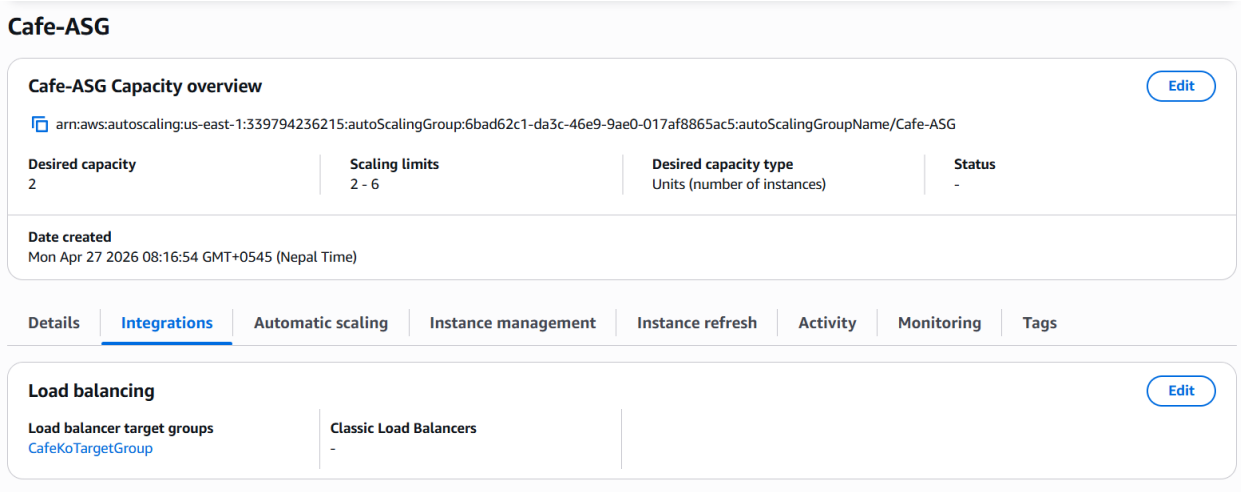
Show two screenshots:

- The EC2 Load Balancers console showing the Application Load Balancer in Active state, with its DNS name visible, placed in the two

public subnets.



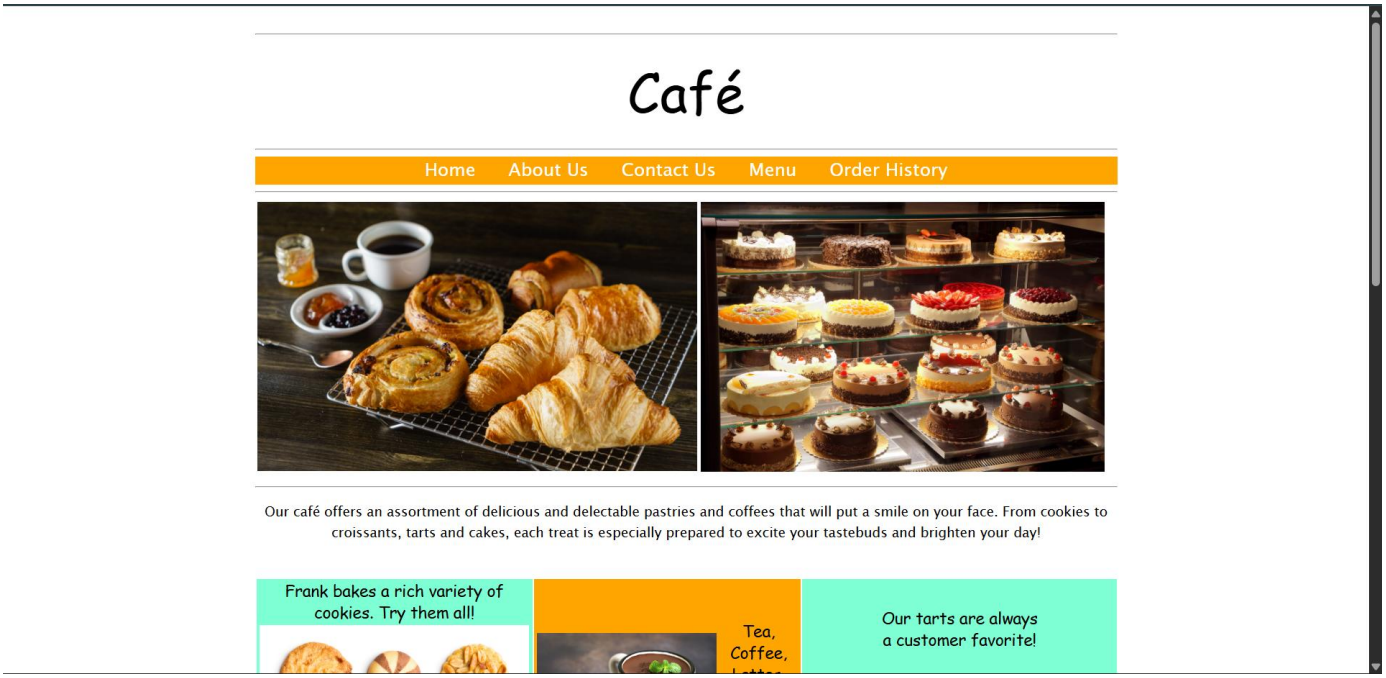
- The Auto Scaling group's Load balancing section (under Details tab) confirming the target group has been attached to the group.



Task 6: Testing load balancing and automatic scaling

Task 6.1: Testing the web application without load

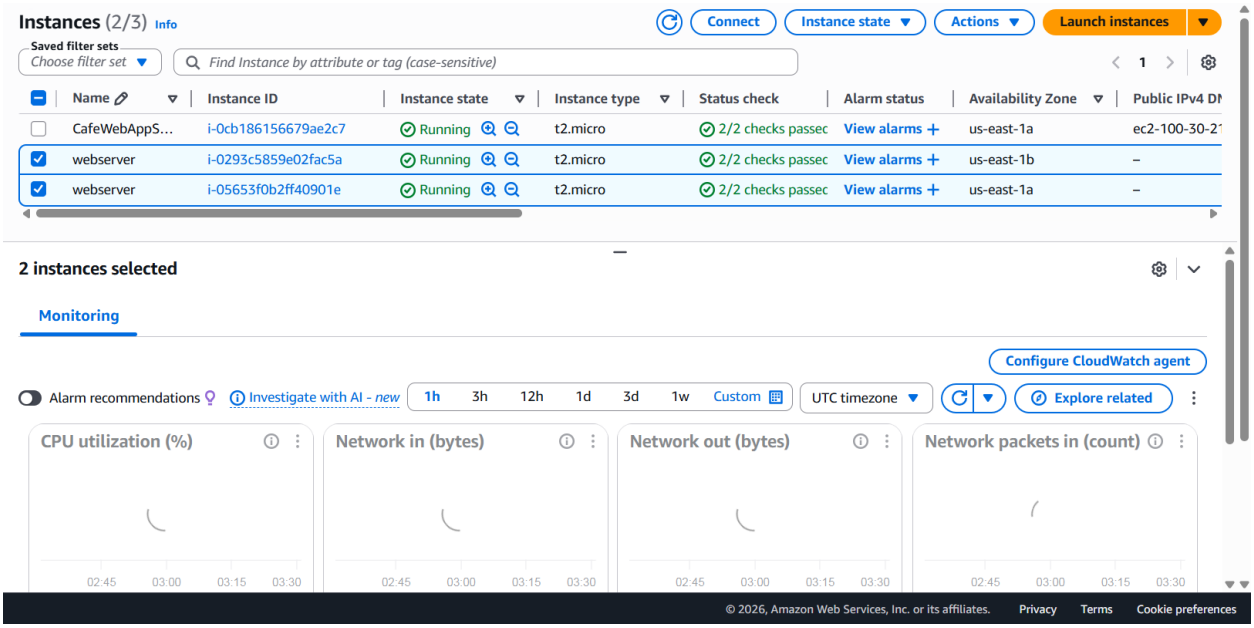
- Show a screenshot of the browser successfully loading the café application via the load balancer DNS name (http://<load-balancer-dns>/cafe), with the full café menu visible.



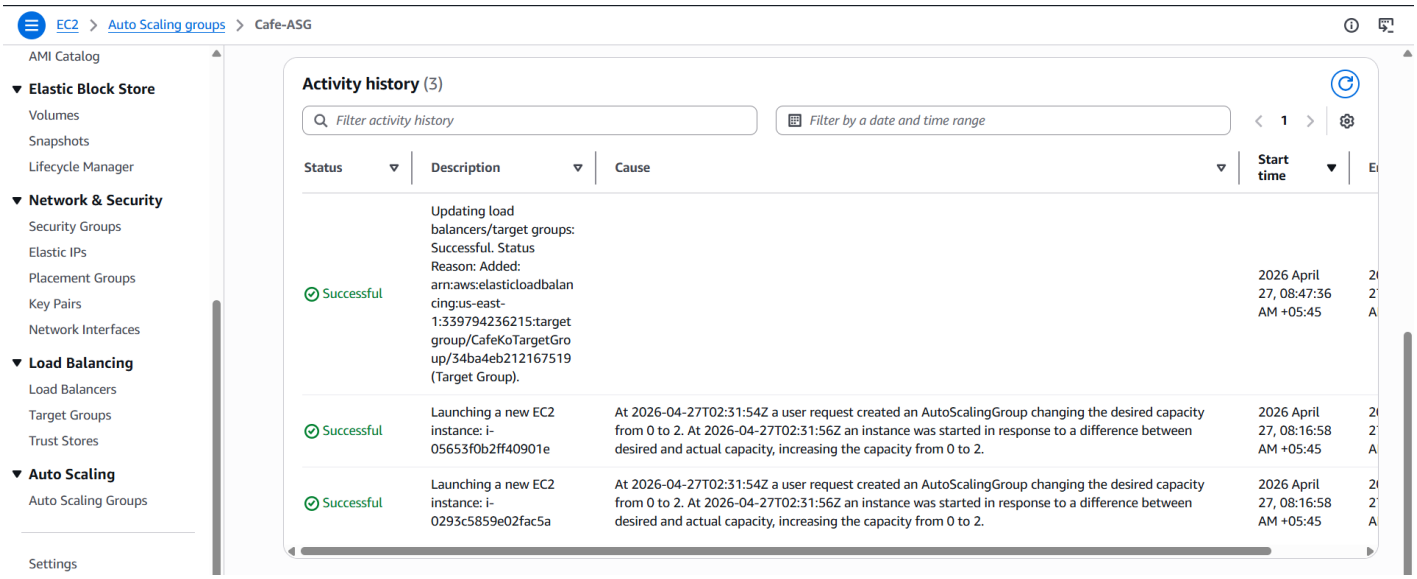
Task 6.2: Testing automatic scaling under load

Show two screenshots taken after running the stress test (stress --cpu 1 --timeout 600) on one of the web server instances:

- **The EC2 console Instances list showing more than two webserver instances in the running state (proof that the Auto Scaling group scaled out in response to the load).**



- **The Auto Scaling group Activity tab showing the scale-out activity history entry, confirming new instances were launched due to the CPU utilization target tracking policy being triggered.**



Description

Write in your own wording and clearly cover all of the following (about 250 to 450 words):

1. what you built and what you did in that lab
2. one security control you used and why it matters
3. one performance or reliability feature that is present and how it helps
4. one issue you had and how you fixed it (or one likely issue and how you would fix it)

Ans:

In this lab, I created a scalable and highly available AWS environment for the café web application. I first inspected the existing VPC, subnets, and the CafeSG security group to understand how the network was configured. I then updated the network to support multiple Availability Zones by creating a NAT gateway in Public Subnet 2 and adding a route from Private Subnet 2 to the new NAT gateway. After that, I created a launch template using the Cafe WebServer Image, t2.micro instance type, CafeSG security group, CafeRole IAM instance profile, and a resource tag with the name webserver. Using this launch template, I created an

Auto Scaling group across Private Subnet 1 and Private Subnet 2 with a desired capacity of 2, minimum capacity of 2, and maximum capacity of 6. I also created an Application Load Balancer in the two public subnets and attached its target group to the Auto Scaling group so that users could access the café application through the load balancer DNS name.

One security control used in this lab was the security group configuration. The CafeSG security group allowed HTTP traffic for the web application, while the web server instances were placed in private subnets instead of being directly exposed to the internet. This matters because it helps control access to the servers and reduces the risk of direct public access to the application instances.

A key performance and reliability feature was the use of the Application Load Balancer together with the Auto Scaling group. The load balancer distributes user requests across healthy web server instances, while Auto Scaling can launch additional instances when CPU usage increases. This helps keep the café website responsive and available during high traffic.

One issue I faced was that the Auto Scaling group automatically increased the desired capacity after the scaling policy was triggered. This caused the submission check to mark the Auto Scaling group as incomplete because it expected the required desired capacity setting. I fixed this by stopping the stress test, allowing

the scaling activity to finish, and setting the desired capacity back to 2 while keeping the minimum at 2 and maximum at 6. After confirming the configuration again, the Auto Scaling group matched the lab requirements.